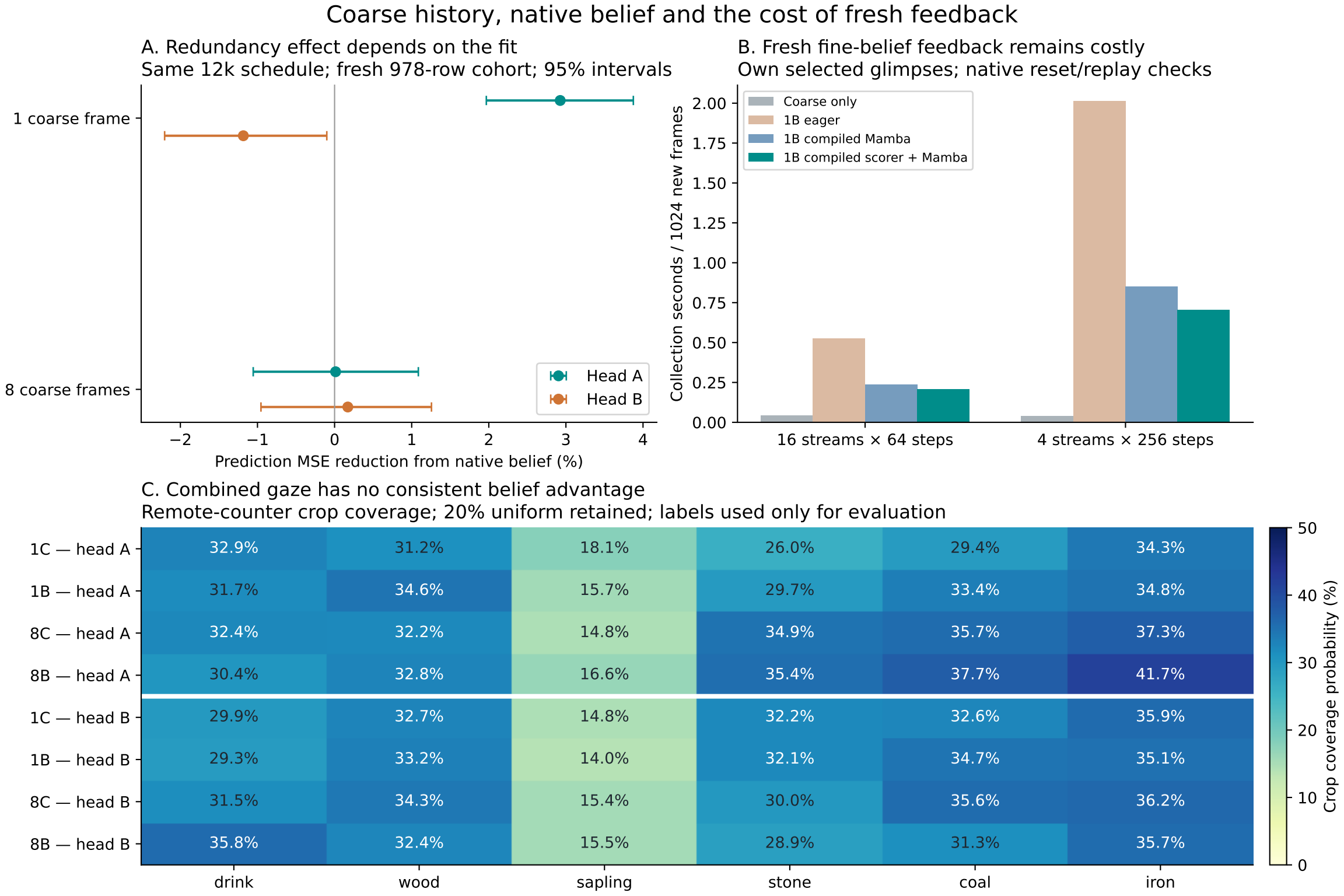
Coarse history, native belief and the cost of fresh feedback
A. Redundancy effect depends on the fit
Same 12k schedule; fresh 978-row cohort; 95% intervals
1 coarse frame
8 coarse frames
Head A
Head B
−2
−1
0
1
2
3
4
Prediction MSE reduction from native belief (%)
B. Fresh fine-belief feedback remains costly
Own selected glimpses; native reset/replay checks
Coarse only
1B eager
1B compiled Mamba
1B compiled scorer + Mamba
Collection seconds / 1024 new frames
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
16 streams × 64 steps
4 streams × 256 steps
C. Combined gaze has no consistent belief advantage
Remote-counter crop coverage; 20% uniform retained; labels used only for evaluation
1C — head A
1B — head A
8C — head A
8B — head A
1C — head B
1B — head B
8C — head B
8B — head B
32.9% 31.2% 18.1% 26.0% 29.4% 34.3%
31.7% 34.6% 15.7% 29.7% 33.4% 34.8%
32.4% 32.2% 14.8% 34.9% 35.7% 37.3%
30.4% 32.8% 16.6% 35.4% 37.7% 41.7%
29.9% 32.7% 14.8% 32.2% 32.6% 35.9%
29.3% 33.2% 14.0% 32.1% 34.7% 35.1%
31.5% 34.3% 15.4% 30.0% 35.6% 36.2%
35.8% 32.4% 15.5% 28.9% 31.3% 35.7%
drink
wood
sapling
stone
coal
iron
Crop coverage probability (%)
0
10
20
30
40
50